

Requirements table :

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall match the EV driver's connector type with compatible available charging sockets and allow the driver to reserve a 30-minute charging window.	High	Pass: The driver sees only compatible connectors and can successfully reserve an available 30-minute slot. Fail: An incompatible connector is shown as reservable or two drivers reserve the same socket for the same time.	Core functionality for finding a compatible charger and preventing reservation conflicts.
FR-002	Functional	The system shall allow the Station Operator to create, modify, enable, disable, and block charging time slots and socket availability.	High	Pass: The operator can update a socket or slot and the revised availability is reflected to drivers. Fail: Drivers can reserve a socket or slot that the operator has blocked.	Allows the operator to manage station capacity and avoid unavailable or maintenance periods.
FR-003	Functional	The system shall display compatible charging stations, available sockets, and the dynamic power rate applicable to the selected charging slot.	High	Pass: For a selected compatible station and slot, the system displays the socket availability and applicable dynamic power rate. Fail: Required availability or rate information is missing or incorrect.	Helps drivers make informed reservation decisions before confirming a slot.
FR-004	Functional	The system shall allow an EV Driver to cancel or modify an existing reservation before the charging slot begins, subject to station rules.	Medium	Pass: An eligible reservation can be cancelled or changed and the updated slot availability is reflected in the system. Fail: A cancelled or changed reservation remains active incorrectly.	Provides flexibility to drivers and allows unused charging capacity to become available to others.
FR-005	Functional	The system shall send a reservation confirmation and charging reminder containing the station, socket, date, time, and reservation details.	High	Pass: After a successful reservation, a confirmation is sent with correct booking details, and a reminder is sent before the reserved slot. Fail: No notification is sent or the details are incorrect.	Reduces missed reservations and gives the driver clear booking information.
NFR-001	Non-Functional — Reliability & Security	The system shall automatically release a locked charging socket and apply the configured penalty if the driver fails to plug in within 10 minutes of the reserved slot start time.	High	Pass: In simulated tests, sockets are released and the configured penalty is applied when no plug-in event occurs within 10 minutes. Fail: The socket remains locked or the required penalty is not processed.	Prevents charging capacity from being blocked by no-shows and supports fair station usage.
NFR-002	Non-Functional — Performance	The system shall display compatible available charging sockets and their dynamic power rates within 2 seconds for at least 95% of requests under the specified peak test load.	High	Pass: Performance testing shows at least 95% of requests complete within 2 seconds. Fail: More than 5% of requests exceed 2 seconds under the specified load.	Ensures drivers can quickly find and reserve charging options, especially during peak demand.